Vaccine 34 (2016) 306–312

Contents lists available at ScienceDirect

Vaccine

journal homepage: www.elsevier.com/locate/vaccine

Optimization of physiological properties of hydroxyapatite as a vaccine adjuvant

Masayuki Hayashi[a,b,c], Taiki Aoshi[a,b,1], Yasumichi Kogai[d], Daisuke Nomi[d], Yasunari Haseda[a,b,1], Etsushi Kuroda[a,b], Kouji Kobiyama[a,b,2], Ken J. Ishii[a,b,*]

[a] Laboratory of Adjuvant Innovation, National Institutes of Biomedical Innovation, Health and Nutrition (NIBIOHN), 7-6-8 Saito-Asagi, Ibaraki, Osaka 567-0085, Japan
[b] Laboratory of Vaccine Science, WPI Immunology Frontier Research Center (iFReC), Osaka University, 3-1 Yamadaoka, Suita, Osaka 565-0871, Japan
[c] Vaccine Research Development Office, Biology Research Laboratories, Sohyaku, Innovative Research Division, Mitsubishi Tanabe Pharma Corporation, 1000 Kamoshida-cho, Aoba-ku, Yokohama 227-0033, Kanagawa, Japan
[d] Technical Development Center, SofSera Corporation, 2-7-1 Ayumino, Izumi, Osaka 594-1157, Japan

ARTICLE INFO

Article history:
Received 25 August 2015
Received in revised form
15 November 2015
Accepted 22 November 2015
Available online 5 December 2015

Keywords:
Adjuvant
Hydroxyapatite
Particles
Vaccine
Size
Calcium phosphate

ABSTRACT

Various particles such as Alum or silica are known to act as an adjuvant if co-administered with vaccine antigens. Several reports have demonstrated that the adjuvanticity is strongly affected by the physicochemical properties of particles such as the size, shape and surface charge, although the required properties and its relationship to the adjuvanticity are still controversial. Hydroxyapatite particle (HAp) composed of calcium phosphate has been shown to work as adjuvant in mice. However, the properties of HAp required for the adjuvanticity have not been fully characterized yet. In this study, we examined the role of size or shape of HAps in the antibody responses after immunization with antigen. HAps whose diameter ranging between 100 and 400 nm provided significantly higher antibody responses than smaller or larger ones. By comparison between sphere and rod shaped HAps, rod shaped HAps induced stronger inflammasome-dependent IL-1β production than the sphere shaped ones *in vitro*. However, sphere- and rod-shaped HAp elicited comparable antibody response in WT mice. Vice versa, *Nlrp3*$^{-/-}$, *Asc*$^{-/-}$ or *Caspase1*$^{-/-}$ mice provided comparable level of antibody responses to HAp adjuvanted vaccination. Collectively, our results demonstrated that the size rather than shape is a more critical property, and IL-1β production *via* NLRP3 inflammasome is dispensable for the adjuvanticity of HAps in mice.
© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

Calcium phosphate is a natural constituent of hard tissue in human body such as bone and teeth, and hydroxyapatite (HA) composed of calcium phosphate is a biocompatible material and used for dental implant materials or bone generation [1,2]. HA implant can be resorbed either by a solution-mediated process (the solubility of implants in physiological solutions) [3] or a cell-mediated process. In particular, macrophages have the ability to degrade HA crystals [4], demonstrating HA is a biodegradable material.

Vaccine is the most effective agent to prevent us from infections, and vaccine adjuvants are used to improve the efficacy for some vaccines. Vaccine adjuvants are classified by different types of compounds such as microbial products, emulsions, particles or liposomes [5–7]. Among these adjuvants, aluminum salts (Alum) is widely used for human vaccines. However, recent studies have suggested some concerns on the safety of Alum, such as the severity of local tissue irritation, longer duration of the inflammatory reaction at the injection site and propensity to elicit undesirable IgE responses [8–10]. Therefore, there is a need for alternative adjuvants. Calcium phosphates (CPs) are one of successfully marketed vaccine adjuvant for human use and has established safety profile [9]. CPs have been shown to improve various vaccines efficacy

* Corresponding author at: Health and Nutrition (NIBIOHN) Innovation 7-6-8 Saito-Asagi, Ibaraki, Osaka 567-0085, Japan. Tel.: +81 72 641 8043; fax: +81 72 641 8079.
E-mail addresses: Hayashi.Masayuki@me.mt-pharma.co.jp (M. Hayashi), aoshi@biken.osaka-u.ac.jp (T. Aoshi), kogai@sofsera.co.jp (Y. Kogai), nomi@sofsera.co.jp (D. Nomi), y-sedaha@biken.osaka-u.ac.jp (Y. Haseda), kuroetu@ifrec.osaka-u.ac.jp (E. Kuroda), kobi@lji.org (K. Kobiyama), kenishii@biken.osaka-u.ac.jp, kenishii@nibiohn.go.jp (K.J. Ishii).
[1] Present address: Vaccine Dynamics Project, BIKEN Innovative Vaccine Research Alliance Laboratories, Research Institute for Microbial Diseases (RIMD), Osaka University, Osaka 565-0871, Japan.
[2] Present address: La Jolla Institute for Allergy and Immunology, La Jolla, CA, USA.

http://dx.doi.org/10.1016/j.vaccine.2015.11.059
0264-410X/© 2015 Elsevier Ltd. All rights reserved.

including diphtheria and tetanus toxoids [11,12]. In addition, CPs have some advantages over the other particle adjuvants. CPs are non-toxic, induce very little inflammation at injection site, have less local tissue irritation and high adjuvanticity without IgE induction [8,9].

The studies of several different types of particle adjuvant have demonstrated that particle size, surface morphology, surface charge and surface area are important parameters that influence on the adjuvanticity [13,14]. In addition, their proinflammatory activities have varied greatly with their physical properties. Rod- or needle-shaped hydroxyapatite (HA) crystals, which resembled the forms of HA aggregates found in the osteoarthritis (OA) synovium, induce higher IL-1β and IL-18 production *via* NLRP3 inflammasome activation in macrophages than sphere-shaped ones, suggesting ectopic deposition of HA crystals in joints induces inflammation closely associated with OA [15]. However the optimal size or shape for the adjuvanticity remains still unclear although nanoparticles of HA composed of CP have shown to have adjuvanticity in various studies [8,16].

In this study, we developed a unique method of synthesis of hydroxyapatite particle (HAp) that the size and shape are desirably controlled. By using several different sized and shaped HAps, we examined the required physicochemical properties and its relationship to the adjuvanticity of HAp in mice.

2. Materials and methods

2.1. Mice

Six-week-old female C57BL/6j mice were purchased from CLEA Japan. *Nlrp3*$^{-/-}$, *Asc*$^{-/-}$ and *Caspase1*$^{-/-}$ mice have been described previously [17]. All animal experiments were performed in accordance with institutional guidelines for the National Institute of Biomedical Innovation, Health and Nutrition animal facility.

2.2. Antigens, antibodies, adjuvants and peptides

Ovalbumin (OVA) was purchased from Seikagaku-kogyo. SplitHA vaccine derived from A/California/7/2009 (H1N1) strain was provided from Institute of Microbial Chemistry (Japan, Osaka). Aluminum hydroxide (Alum) was purchased from Invivogen. Cytokine ELISA kits for IL-1β and TNFα were purchased from R&D systems.

2.3. Hydroxyapatite particles (HAps)

Preparation of HAps were described previously [18,19], with some modifications. Briefly, HAps synthesized by previous method were sterilized by dry heat treatment at 300 °C for 2 h in a pre-sterilized glass ample. After the ample was sealed, dry heat treatment was repeated again.

2.4. Scanning electron microscopy (SEM) analysis

The size and shape of HAps were observed by SEM (JSM-6301F; JEOL Ltd., Japan, Tokyo).

2.5. Immunization

C57BL/6 J mice were immunized twice i.d. (at the tailbase) with 2 weeks intervals (day 0 and 14). For antigen-specific ELISA, blood samples were taken on day 14 and 28. For During vaccination and bleeding, the mice were anesthetized with ketamine. Antigen-loaded Alum or HAp was rotated for more than 1 h before the immunization. Alum was used at 0.67 mg/mouse for immunization.

2.6. Ab titer

For ELISA, 96 well plates were coated with 1 μg/ml SplitHA in a carbonate buffer (pH9.6) for SplitHA-vaccinated groups or 10 μg/ml OVA for OVA-vaccinated groups. Wells were blocked with PBS containing 1% BSA and diluted sera from immunized mice were incubated on the antigen-coated plate. After washing, goat anti-mouse total IgG, IgG1 or IgG2c conjugated HRP (Southern Biotech) were added and incubated for 1 h at RT. After additional washing, the plates were incubated with TMB for 30 min and the reaction was stopped with 1 N H_2SO_4 and then the absorbance was measured. Ab titer was calculated. OD of 0.2 was set as the cut-off value for positive samples.

2.7. Virus neutralization titer

The sera from immunized mice were mixed with RDE II (Denka-Seiken) and incubated for overnight at 37 °C. After further incubation for 1 h at 56 °C, serially diluted sera and a final infecting titer of 100 $TCID_{50}$/ml of A/California/7/2009 (H1N1) influenza virus in MEM media containing 10 mM HEPES, 1% Penicillin-Streptomycin, 0.2% bovine albumin and 10 μg/ml trypsin were incubated for 30 min at 37 °C. After that, these were added to Madin–Darby Canine kidney (MDCK) cells. Four days after incubation at 37 °C with 5% CO_2, the MDCK cells were fixed with 10% formalin for 10 min at room temperature. The cells were stained with Naphtol Blue Black Solution (Naphtol Blue Black 0.5 g, Sodium Acetate 0.5 g, Acetic Acid 45 ml and distilled water 455 ml) for 30 min at room temperature. The stained cells were washed well with water and then dried. Subsequently, 0.1 M NaOH was added to the cells and the plates were read at 630 nm in a microplate reader. Virus neutralization titer was determined by maximum dilution ratio which gave higher absorbance than average of positive and negative controls.

2.8. In vitro stimulation of macrophages

For macrophages preparation, mice were i.p. injected with 3 ml of 4% (w/v) thioglycolate (SIGMA) solution. Four days later, macrophages were collected from peritoneal cavity and plated on 96 well plates. Macrophages were primed with 50 ng/ml LPS for 15 h, and stimulated with adjuvants for 8 h. IL-1β or TNFα in supernatants was measured by ELISA.

2.9. Statistical analysis

Statistical significance ($P < 0.05$) between groups was determined by Dunnet's Multiple comparison test or student *t*-test.

3. Results

3.1. Several hundred nanometer HAps have a better adjuvanticity than other larger or smaller ones

To investigate the relationship between particle size and adjuvanticity of HAp, we first prepared sphere-shaped HAps with sizes ranging from nanometer to micrometer including 40 nm (S40), 100 nm (S100), 170 nm (S170), 400 nm (S400), 1.8 μm (S1800) and 5 μm (S5000) on average (Table 1 and Fig. 1A–F). C57BL/6j mice were immunized twice with influenza splitHA vaccine (SV) plus three different doses (0.2, 1 or 5 mg) of these HAps, and the antigen-specific antibody response was examined. One hundred to 400 nm particles induced higher antibody response than smaller (40 nm) and larger (1.8 μm or 5 μm) sized particles (Fig. 2A–C). IgG subclass analysis also revealed that HAp induced more IgG1

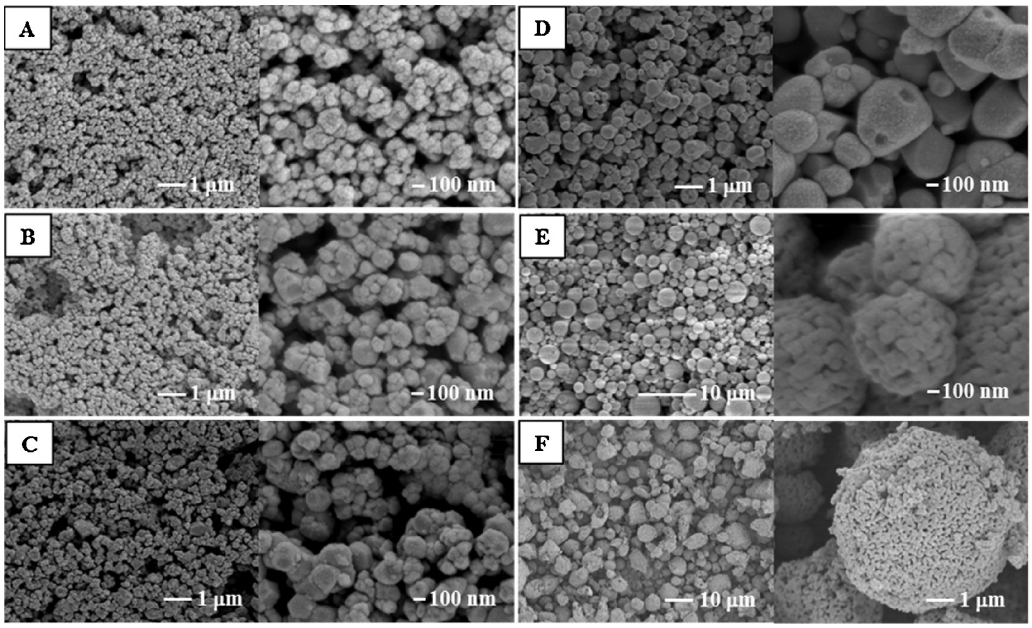

Fig. 1. *Various sized sphere shape HAps.* Scanning electron micrographs of (A) S40, (B) S100, (C) S170, (D) S400, (E) S1800 and (F) S5000. (Left) lower and (right) higher magnified SEM photographs of HAp. Pictures were taken under (A–D) 10,000× or 40,000×; (E) 2000× or 40,000×; (F) 1000× or 10,000× magnifications in the left or right, respectively.

Table 1
The size and shape of HAps. HAps size was determined by mean value of diameter of 100 or more HAps on a picture taken by SEM at magnification of 40,000. The shape, size and standard deviation (SD) were shown for each HAps.

Name	Shape	Size (nm)	SD
S40	Sphere	41	32
S100	Sphere	104	74
S170	Sphere	172	98
S400	Sphere	396	132
S1800	Sphere	1796	1093
S5000	Sphere	5191	3266
R120	Rod	123	59
R160	Rod	157	82
R250	Rod	253	136

than IgG2c production, suggesting that HAps generate Th2-biased immune response (Fig. 2B and C). Furthermore, SV plus 100 to 400 nm particles immunization showed higher virus neutralization titers (Fig. 2D). These results demonstrated that several hundred nanometer sizes of sphere-shaped HAps have a better adjuvanticity than other larger or smaller ones.

3.2. Rod-shaped HAps also have an adjuvanticity

We also prepared rod-shaped HAps including 120 (R120), 160 (R160), and 250 nm (R250) size of HAp (Fig. 3 and Table 1). Similar to the sphere-shaped HAp, C57BL/6j mice were immunized twice with SV plus these rod-shaped HAps, and the antigen-specific antibody response was determined. These rod-shaped HAps also induced significantly higher total IgG, IgG1 (Fig. 4A–C), and virus neutralization titers (Fig. 4D) than SV alone immunization, suggesting that the rod-shaped HAp also showed adjuvanticity and induced Th2-type responses to the SV vaccine. Collectively, optimal size of HAp as vaccine adjuvant is around several hundred nanometers, and the both sphere and rod shapes have similar adjuvanticity to the SV vaccine.

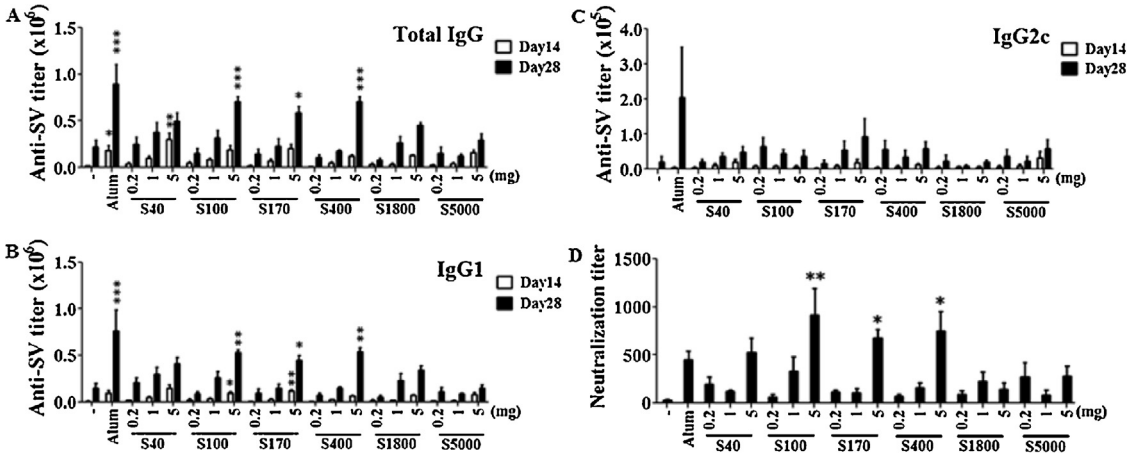

Fig. 2. *Several hundred nanometer HAps induce higher antibody response.* On day 0 and 14, mice ($n = 5$) were immunized i.d. with 1 μg SV alone or together with adjuvant. Antigen-specific (A) total IgG, (B) IgG1 and (C) IgG2c titers in sera at day 14 and 28 were measured by ELISA. (D) Viral neutralization titer in sera at day 28 was measured by microneutralization assay. Results presented representatives of three separate experiments. Median and SEM are shown for each group. Statistical significances are indicated, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ on Dunnet's Multiple comparison test.

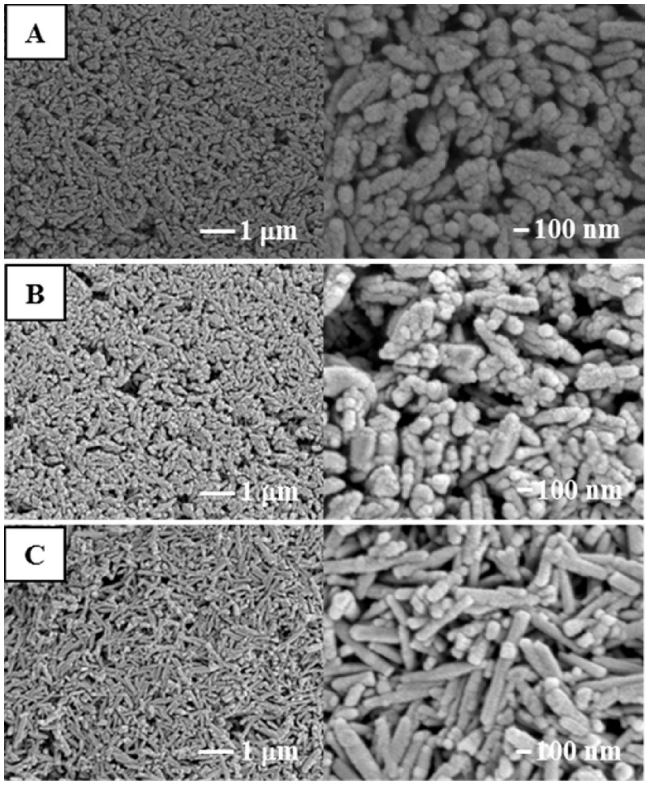

Fig. 3. *Various sized rod shape HAps.* Scanning electron micrographs of (A) R120, (B) R160 and (C) R250. Pictures were taken under 10,000× or 40,000× magnifications in the left or right, respectively.

3.3. Rod-shaped HAps induce higher IL-1β production in macrophages

Several reports have demonstrated that particles such as silica or Alum, stimulate macrophages and dendritic cells to produce IL-1β through the activation of NLRP3 inflammasome. In addition, recent reports showed that rod- or needle-shaped HA crystal has shown to stimulate robust secretion of proinflammatory cytokines such as IL-1β and IL-18 from macrophages in a NLRP3 inflammasome-dependent manner, whereas sphere-shaped HA crystal provided minimal effect on these cytokines induction [15]. Therefore, to examine the ability of HAp to produce IL-1β, thioglycolate-induced peritoneal macrophages were stimulated with sphere or rod shaped HAps with LPS priming. As reported, rod-shaped HAp induced higher IL-1β production than sphere-shaped HAp (Fig. 5A). In addition, larger HAp (S5000) that showed weaker adjuvanticity (Fig. 2), did not induce significant IL-1β production. We further examined the involvement of inflammasome in the IL-1β production by using peritoneal macrophages derived from *Caspase1*$^{-/-}$ mice. Caspase1 is an essential component of inflammasome which cleaves pro-IL-1β to mature form. As observed in Alum, the ability of HAps to induce IL-1β was completely dependent on Caspase1 (Fig. 5B), whereas TNFα production induced by LPS was not affected by *Caspase1* deficiency (Supplementary Fig. 1A–B), suggesting that HAps induce IL-1β production in an inflammasome dependent manner.

3.4. NLRP3 inflammasome activation is not required for the adjuvanticity of HAps

Finally, we investigated the requirement of NLRP3 inflammasome activation for the HAps adjuvanticity *in vivo* by immunizing *Nlrp3*$^{-/-}$, *Asc*$^{-/-}$ or *Caspase1*$^{-/-}$ mice. For this experiment, we selected S170 and R160 as each representative from spheres and rods of HAps. Mice lacking *Nlrp3*, *Asc* or *Caspase1* were i.d. immunized twice with OVA-loaded S170, R160, or Alum, and OVA-specific antibody response was examined. All knockout mice we tested showed comparable amount of OVA specific antibody titer (Fig. 6A–C), in spite of the complete IL-1β suppression from macrophage *in vitro* in the absence of *Caspase1* (Fig. 5A and B). These results demonstrated that the adjuvanticity of HAps is not dependent on the NLRP3 inflammasome activation *in vivo*.

4. Discussion

Various adjuvants based on nanoparticles or microparticles are being developed. These particulates enhance cellular uptake

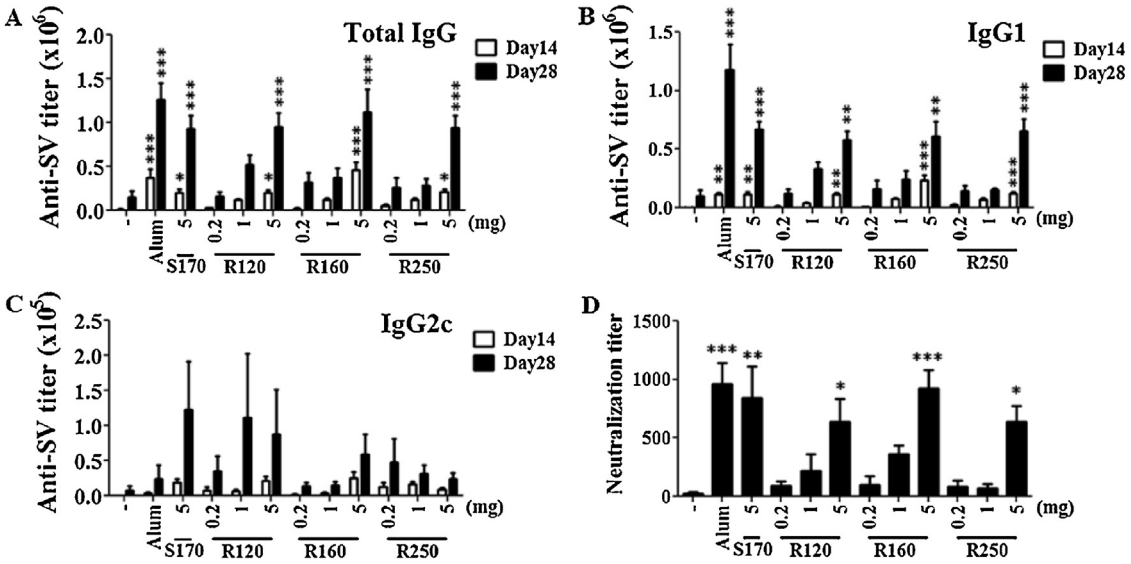

Fig. 4. *Rod-shaped HAps also have an adjuvanticity.* On day 0 and 14, mice ($n = 5$) were immunized i.d. with 1 μg SV alone or together with adjuvant. Antigen-specific (A) total IgG, (B) IgG1 and (C) IgG2c titers in sera at day 14 and 28 were measured by ELISA. (D) Viral neutralization titer in sera at day 28 was measured by microneutralization assay. Results presented representatives of three separate experiments. Median and SEM are shown for each group. Statistical significances are indicated, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ on Dunnet's Multiple comparison test.

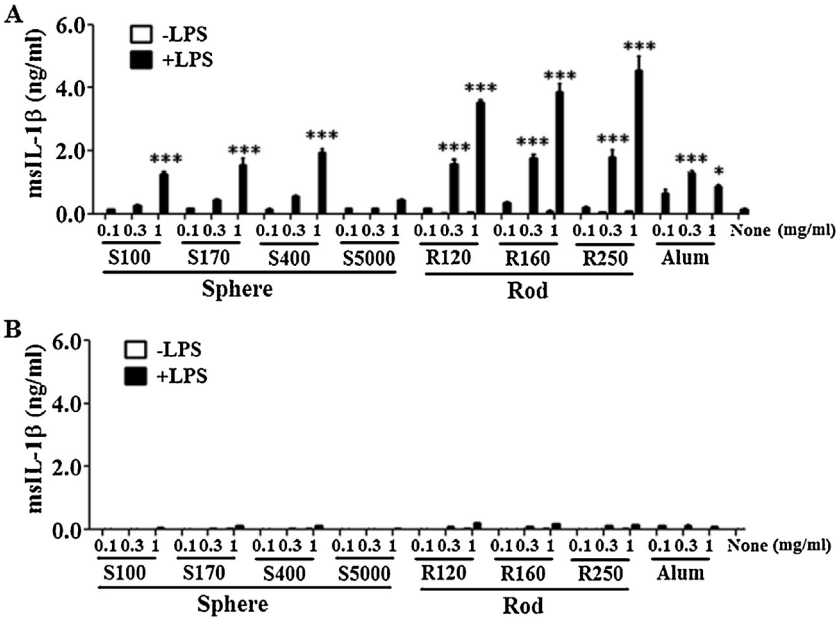

Fig. 5. *Several hundred nanometer HAps induce IL-1β production via inflammasome in macrophages in vitro.* Peritoneal macrophages prepared from (A) C57BL/6j or (B) *Caspase1*$^{-/-}$ mice were primed with 50 ng/ml LPS for 15 h and then stimulated with HAp or Alum at indicated concentration for 8 h. IL-1β secreted into culture supernatants was measured by ELISA. Results presented representatives of three separate experiments. Median and SEM are shown for each group. Statistical significances compared to non-stimulation group are indicated, *** $P < 0.001$ on Dunnet's Multiple comparison test.

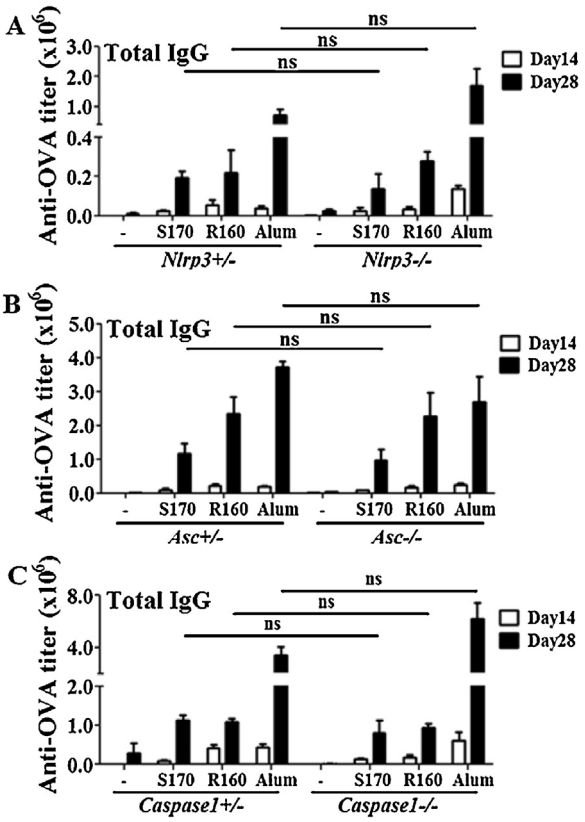

Fig. 6. *NLRP3 inflammasome activation is not required for the adjuvanticity of HAps.* On day 0 and 14, (A) *Nlrp3*$^{-/-}$, (B) *Asc*$^{-/-}$ or (C) *Caspase1*$^{-/-}$ mice ($n = 3$–6) were immunized i.d. with 10 μg OVA alone or together with adjuvant. S170 and R160 were used at 5 mg per mouse. Antigen-specific total IgG titers in sera at day 14 and 28 were measured by ELISA. Results presented representatives of two separate experiments. Median and SEM are shown for each group. Statistical significances are determined by Student *t*-test.

of co-administered antigen and antigen presentation. However, non-biodegradable particles such as alum, widely used for human vaccines, are retained for long time and sometimes cause induration at the injected site. Thus, biodegradable particles have an advantage over non-biodegradable particles to be less toxic [20]. HAp which is a main component of bone or teeth is a biodegradable and biocompatible material. In fact, HAps have been shown to be solubilized in acidic or basic media [3] and to be degraded by macrophages.

For particle adjuvants, understanding the influence of the physicochemical properties of particulate on immunity is important to design and develop effective adjuvants. It has been reported that the size, surface charge, surface hydrophilicity/lipophilicity and antigen-adjuvant binding strength obviously influence on the adjuvant effect of particles [21–23]. In this study, we focused to analyze the relationship of the size or shape and adjuvanticity of HAps. Generally, particulates with a comparable size to pathogens such as virus are easily recognized and taken up by dendritic cells (DCs) or macrophages [23]. Small particles (20–200 nm) are preferentially taken up *via* endocytosis by DCs and larger particles (<5 μm) are phagocytosed by macrophages, leading to efficient co-administered antigen presentation and subsequent adaptive immune response [24]. This size-dependent immunity has been shown by various particulates. For instance, tetanus toxoid-loaded sulfobutylated poly (vunyl alcohol)-graft-PLGA or silica nanoparticles provided higher antibody response than their microparticles [25,26]. In comparison of immune response induced by ovalbumin (OVA)-conjugated polystyrene beads with various sizes (20, 40, 100 and 500 nm, and 1 and 2 μm), 40 nm induced strongest OVA-specific antibody response [27]. Hemozoin particles with sizes in the range of 50–200 nm exhibited a stronger adjuvant effect than larger (2–20 μm) and smaller (<50 nm) particles [28]. These reports have suggested that nanometer size is preferable to micron size for the adjuvant activity. On the contrary, bovine serum albumin (BSA)-entrapped in poly (d,l-lactic-co-glycolic) (PLGA) or hepatitis B surface antigen-loaded polylactic acid (PLA) microparticles induce higher serum IgG antibody response than their nanoparticles [29,30], suggesting micrometer size is better than nanometer

size. Collectively, there seems to be optimal size range for each particulate adjuvant. In our study, the adjuvanticity of different sized HAps ranging 40 nm to 5 μm were compared (Fig. 2). Our finding demonstrated that HAps showed the size-dependent immunity, consistent with previous reports of particulates composed of other materials [21,23,24], and 100 to 400 nm is optimal size range for the adjuvanticity of HAps. Also, the shape-dependent immunity of particulates has been shown by gold nanoparticles with spherical, rod or cubic shape [31]. In this study, spherical shaped gold nanoparticles induce highest antigen-specific antibody response in these particles. However, rod and spherical shaped HAps with similar size provided comparable antibody response, demonstrating the shape is not essential for the adjuvanticity of HAps in our study (Fig. 4).

Recently, the necessity of attachment of antigen to particulates for the adjuvanticity has been questioned [26] even though antigen absorbed, entrapped or conjugated particulates have shown high immunogenicity in various studies [8,32,33]. Wibowo et al. has shown that antibody response is significantly enhanced by silica nanoparticles without specific conjugation of antigen [26]. Additionally, inulin-derived microparticle, Advax^{TM} shows the adjuvant effect when administered 24 h prior to the administration of antigen, demonstrating the association of antigen with particles is not required for the adjuvant effect [34]. On the other hand, typical particle adjuvant, Alum does not show the adjuvant effect without pre-absorption and co-administration with antigen [34]. These results have suggested that the association of antigen and particulate such as absorption or conjugation is not always necessary for the adjuvant effect of particles. Most attachments of antigen to particles are achieved through multiple steps of chemical reaction or physical stress that sometime changes antigen conformation or increases formulation toxicity [26,35,36]. Therefore, particulates that do not require the association with antigen for the adjuvanticity are superior vaccine adjuvant to other particulates. Although antigen absorbed HAps were used in this study, the necessity of association antigen with HAps still remains unclear.

The mechanism of action of particle adjuvants is considered as depot effect, antigen uptake into antigen presenting cells, and activation of innate immunity. Some reports have demonstrated that NLRP3 inflammasome activation is essential for adjuvanticity of Alum [37,38]. Since other particle adjuvants also activate NLRP3 inflammasome *in vitro*, NLRP3 inflammasome activation is considered to be required for their adjuvanticity. However, other reports have also shown that the adjuvanticity of particles including Alum is independent of NLRP3 inflammasome [28,39,40]. Therefore, we examined the involvement of NLRP3 inflammasome in the adjuvanticity of HAps. Consistent with previous study [15], HAps generated proinflammatory cytokine IL-1β production *via* NLRP3 inflammasome from macrophages *in vitro*. Especially, rod-shaped HAps provided higher IL-1β production than sphere-shaped HAps, implying the shape-dependent response (Fig. 5). However, rod shaped HAps induced comparable antibody response to sphered HAps (Fig. 4). In addition, absence of NLRP3 inflammasome component, *Nlrp3*, *Asc* or *Caspase1* provided significant antibody response *in vivo* even though IL-1β production induced by HAps was completely suppressed in the absence of *Caspase1 in vitro* (Figs. 5 and 6). Of note, the adjuvanticity of Alum was not affected by the absence of *Nlrp3*, *Asc* or *Caspase1* in our study. These results revealed that the ability to produce IL-1β *via* NLRP3 inflammasome activation is not required for the adjuvanticity of both particles, HAps and Alum.

Together, our results suggested that the size range is a primary factors affecting and controlling the adjuvanticity of the particles *in vivo*. Although innate immune signaling is usually essential for the induction of adaptive immune response, our results suggested that at least, NLRP3 inflammasome dependent IL-1β production is not the absolute requirement of particulate adjuvants such as Alum and HAp. These findings provided important implications for designing future particle adjuvant developments.

Conflict of interest statement

The authors have no financial conflicts of interest.

Acknowledgements

We thank Akiko Okabe, Mariko Nakamura and Kousaku Murase for excellent technical assistance with animal husbandry and genotyping, as well as the members of K.J.I.'s laboratories for valuable comments and helps.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.vaccine.2015.11.059.

References

[1] Furuhashi K, Akasaka T, Kitagawa Y, Watari F. Evaluation of adhesion between material and epithelium using a three-dimensional human epidermal model. Nano Biomed 2012;4(2):76–84.
[2] Hatakeyama W, Taira M, Takafuji K, Kihara H, Kondo H. Bone-regeneration trial of rat critical-size calvarial defects using nano-apatite/collagen composites. Nano Biomed 2013;5(2):95–103.
[3] Jarcho M. Calcium Phosphate Ceramics as Hard Tissue Prosthetics. Clin Orthop Relat Res 1981;157:259–78.
[4] Kwong Cheung Hing, Burns WB, Cheng HS. Solubilization of hydroqyapatite crystals by murine bone cells, macrophages and fibroblasts. Biomaterials 1989;10:579–84.
[5] Tritto E, Mosca F, De Gregorio E. Mechanism of action of licensed vaccine adjuvants. Vaccine 2009;27(25–26):3331–4.
[6] Brito LA, Malyala P, O'Hagan DT. Vaccine adjuvant formulations: a pharmaceutical perspective. Semin Immunol 2013;25(2):130–45.
[7] Hedayat M, Takeda K, Rezaei N. Prophylactic and therapeutic implications of toll-like receptor ligands. Med Res Rev 2012;32(2):294–325.
[8] He Q, Mitchell AR, Johnson SL, Wagner-Bartak C, Morcol T, Bell SJD. Calcium phosphate nanoparticle adjuvant. Clin Diagn Lab Immunol 2000;7(6):899.
[9] Tchavdar LV. Aluminium phosphate but not calcium phosphate stimulates the specific IgE response in guinea pigs to tetanus toxoid. Allergy 1978;33:155–9.
[10] Goto Norihisa, Kate Hiroshi, Maeyama Jun-ichi, Shibano Michiko, Saito Takuya, Yamaguchit Junji, et al. Local tissue irritating effects and adjuvant activities of calcium phosphate and aluminium hydroxide with different physical properties. Vaccine 1997;15(12/13):1364–71.
[11] Aggerbeck H, Heron I. Adjuvanticity of aluminium hydroxide and calcium phosphate in diphtheria-tetanus vaccines-I. Vaccine 1995;13(14):1360–5.
[12] Aggerbeck H, Fenger C, Heron I. Booster vaccination against diphtheria and tetanus in man. Comparison of calcium phosphate and aluminium hydroxide as adjuvants-II. Vaccine 1995;13(14):1366–74.
[13] Kuroda E, Coban C, Ishii KJ. Particulate adjuvant and innate immunity: Past Achievements. Present findings, and future prospects. Int Rev Immunol 2013;32(2):209–20.
[14] Jiang D, Premachandra GS, Johnston C, Hem SL. Structure and adsorption properties of commercial calcium phosphate adjuvant. Vaccine 2004;23(5):693–8.
[15] Jin C, Frayssinet P, Pelker R, Cwirka D, Hu B, Vignery A, et al. NLRP3 inflammasome plays a critical role in the pathogenesis of hydroxyapatite-associated arthropathy. Proc Natl Acad Sci U S A 2011;108(36):14867–72.
[16] He Q, Mitchell A, Morcol T, Bell SJD. Calcium phosphate nanoparticles induce mucosal immunity and protection against herpes simplex virus type 2. Clin Vaccine Immunol 2002;9(5):1021–4.
[17] Onishi M, Ozasa K, Kobiyama K, Ohata K, Kitano M, Taniguchi K, et al. Hydroxypropyl-beta-cyclodextrin spikes local inflammation that induces Th2 cell and T follicular helper cell responses to the coadministered antigen. J Immunol 2015;194(6):2673–82.
[18] Okada M, Furuzono T. Nano-sized ceramic particles of hydroxyapatite calcined with an anti-sintering agent. J Nanosci Nanotechnol 2007;7(3):848–51.
[19] Okada M, Furuzono T. Calcination of rod-like hydroxyapatite nanocrystals with an anti-sintering agent surrounding the crystals. J Nanopart Res 2006;9(5):807–15.
[20] Ahmed KK, Geary SM, Salem AK. Applying biodegradable particles to enhance cancer vaccine efficacy. Immunol Res 2014;59(1–3):220–8.
[21] Yan S, Gu W, Xu ZP. Re-considering how particle size and other properties of antigen-adjuvant complexes impact on the immune responses. J Colloid Interface Sci 2013;395:1–10.

[22] Oyewumi MO, Kumar A, Cui Z. Nano-microparticles as immune adjuvants: correlating particle sizes and the resultant immune responses. Expert Rev Vaccines 2010;9(9):1095–107.
[23] Zhao L, Seth A, Wibowo N, Zhao CX, Mitter N, Yu C, et al. Nanoparticle vaccines. Vaccine 2014;32(3):327–37.
[24] Shah Ruchi R, O'Hagan DT, Amiji Mansoor M, Brito Luis A. The impact of size on particulate vaccine adjuvants. Nanomedicine (Lond) 2014;9(17):2671–81.
[25] Jung T, Kamm W, Breitenbach A, Hungerer KD, Hundt E, Kissel T. Tetanus toxoid loaded nanoparticles from sulfobutylated poly(vinyl alcohol)-graft-poly(lactide-*co*-glycolide): evaluation of antibody response after oral and nasal application in mice. Pharm Res 2001;18(3):352–60.
[26] Wibowo N, Chuan YP, Seth A, Cordoba Y, Lua LH, Middelberg AP. Co-administration of non-carrier nanoparticles boosts antigen immune response without requiring protein conjugation. Vaccine 2014;32(29):3664–9.
[27] Fifis T, Gamvrellis A, Crimeen-Irwin B, Pietersz GA, Li J, Mottram PL, et al. Size-dependent immunogenicity: therapeutic and protective properties of nano-vaccines against tumors. J Immunol 2004;173(5):3148–54.
[28] Coban C, Igari Y, Yagi M, Reimer T, Koyama S, Aoshi T, et al. Immunogenicity of whole-parasite vaccines against *Plasmodium falciparum* involves malarial hemozoin and host TLR9. Cell Host Microbe 2010;7(1):50–61.
[29] Gutierro I, Hernández RM, Igartua M, Gascón AR, Pedraz JL. Size dependent immune response after subcutaneous, oral and intranasal administration of BSA loaded nanospheres. Vaccine 2002;21:67–77.
[30] Kanchan V, Panda AK. Interactions of antigen-loaded polylactide particles with macrophages and their correlation with the immune response. Biomaterials 2007;28(35):5344–57.
[31] Niikura Kenichi, Matsunaga T, Suzuki Tadaki, Kobayashi Shintaro, Yamaguchi Hiroki, Orba Yasuko, et al. Gold nanoparticles as a vaccine platform: influence of size and shape on immunological responses in vitro and in vivo. ACS Nano 2013;7(5):3926–38.
[32] Wendorf J, Singh M, Chesko J, Kazzaz J, Soewanan E, Ugozzoli M, et al. A practical approach to the use of nanoparticles for vaccine delivery. J Pharm Sci 2006;95(12):2738–50.
[33] Slutter B, Soema PC, Ding Z, Verheul R, Hennink W, Jiskoot W. Conjugation of ovalbumin to trimethyl chitosan improves immunogenicity of the antigen. J Control Release 2010;143(2):207–14.
[34] Saade F, Honda-Okubo Y, Trec S, Petrovsky N. A novel hepatitis B vaccine containing Advax^{TM}, a polysaccharide adjuvant derived from delta inulin, induces robust humoral and cellular immunity with minimal reactogenicity in preclinical testing. Vaccine 2013;31(15):1999–2007.
[35] Kunzmann A, Andersson B, Thurnherr T, Krug H, Scheynius A, Fadeel B. Toxicology of engineered nanomaterials: focus on biocompatibility, biodistribution and biodegradation. Biochim Biophys Acta 2011;1810(3):361–73.
[36] Mcneil SE, Dobrovolskaia MA. Immunological properties of engineered nanomaterials. Nat Nanotechnol 2007;2:469–78.
[37] Eisenbarth SC, Colegio OR, O'Connor W, Sutterwala FS, Flavell RA. Crucial role for the Nalp3 inflammasome in the immunostimulatory properties of aluminium adjuvants. Nature 2008;453(7198):1122–6.
[38] Li H, Willingham SB, Ting JPY, Re F. Cutting edge: inflammasome activation by alum and alum's adjuvant effect are mediated by NLRP3. J Immunol 2008;181(1):17–21.
[39] McKee AS, Munks MW, MacLeod MK, Fleenor CJ, Van Rooijen N, Kappler JW, et al. Alum induces innate immune responses through macrophage and mast cell sensors, but these sensors are not required for alum to act as an adjuvant for specific immunity. J Immunol 2009;183(7):4403–14.
[40] Franchi L, Nunez G. The Nlrp3 inflammasome is critical for aluminium hydroxide-mediated IL-1beta secretion but dispensable for adjuvant activity. Eur J Immunol 2008;38(8):2085–9.